

ConfusionLens: Dynamic and Interactive Visualization for Performance Analysis of Multiclass Image Classifiers

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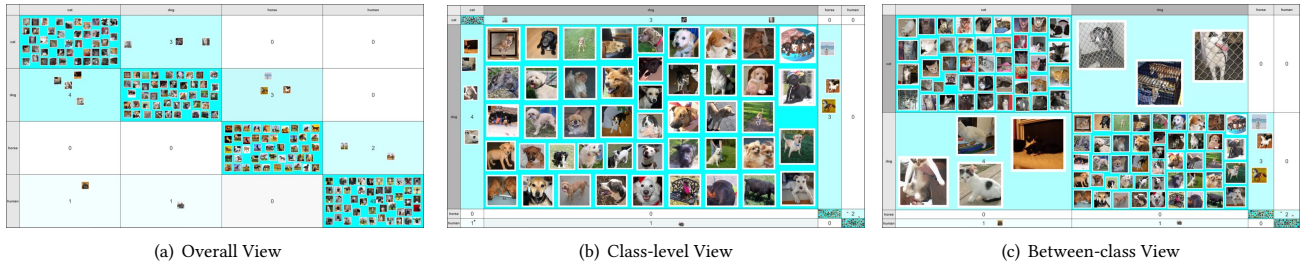


Figure 1: ConfusionLens is a novel visualization interface for performance analysis of multiclass image classifiers that allows users to seamlessly switch layouts among three views (a-c) while observing all of the instances in a single screen.

ABSTRACT

Building higher-quality image classification models requires better performance analysis (PA) methods to help understand their behaviors. We propose ConfusionLens, a dynamic and interactive visualization interface that augments a conventional confusion matrix with focus+context visualization. This interface makes it possible to adaptively provide relevant information for different kinds of PA tasks. Specifically, it allows users to seamlessly switch table layouts among three views (overall view, class-level view, and between-class view) while observing all of the instance images in a single screen. This paper presents a ConfusionLens prototype that supports hundreds of instances and its several extensions to further support practical PA tasks, such as activation map visualization and instance sorting/filtering.

CCS CONCEPTS

• **Human-centered computing** → **Visual analytics**; *Graphical user interfaces*; • **Computing methodologies** → *Machine learning*.

KEYWORDS

User Interface, Image Classification, Confusion Matrices, Performance Analysis, Interactive Visualization

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1 INTRODUCTION

Image classification is one of the major tasks in machine learning (ML) for modeling various problems. However, due to the massive parameter space and complex architectures, many of the classifiers have essentially become black-box systems, increasing the difficulty for model developers to understand, diagnose, and explain model behaviors. Therefore, performance analysis (PA) has become nearly essential to understand and improve a classification model, and various visualization tools supporting PA are being explored [3, 5].

Traditionally, confusion matrices, have frequently been used as one of the PA tasks. This approach has the advantages of intuitiveness and universality in terms of visualizing class-level performance. However, when evaluating a model with an incomparable structure or a similar final score to another model, the aggregated values in the matrices are unable to provide an internal analysis of model behavior. Therefore, we believe integrating existing confusion matrices into instance-level PA techniques to gain insight

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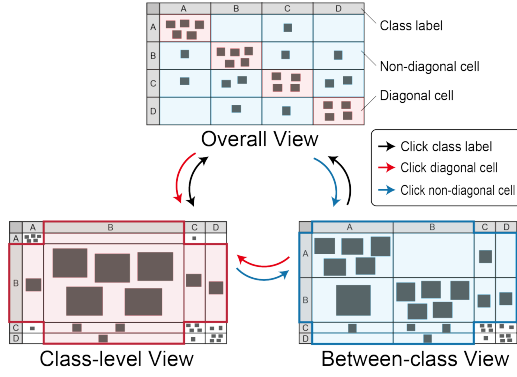


Figure 2: View transitions among the three views. Each transition is accomplished with a single click.

into model behavior would be beneficial to effectively visualize important information (e.g., prediction score distribution, cause of misclassification). Although several work have attempted to augment confusion matrices [1, 7], visualizing instances have not been studied.

We propose ConfusionLens, a novel visualization interface that supports a wide range of PA tasks in multiclass image classification. The core idea is that it applies focus+context visualization techniques to a conventional confusion matrix, thereby enabling users to observe information at different levels of detail in a single screen (i.e., without splitting the screen). This achieves a seamless transition between different PA tasks.

2 CONFUSIONLENS

2.1 Key design

ConfusionLens augments confusion matrices by arranging instances directly on the table to support instance-level observation. In addition, the interface employs a table-based focus+context technique (as seen in prior work [4, 7]) for its layout. With this technique, fish-eye effects can be applied to selectively expand rows and columns to seamlessly switch table layouts among three views: overall view, class-level view, and between-class view. These views can be simply transited by clicking (or tapping) each of the cells or class labels (Fig. 2). Furthermore, we implemented two extensions that support practical PA tasks (as described in Section 2.3).

2.2 Three views

2.2.1 Overall View. Overall view basically follows regular confusion matrices showing the ground truth of class in the horizontal axis and the prediction of class in the vertical axis. Furthermore, we add instances inside the cells and make the number overlaying the instances translucent (Fig. 1(a) shows its example). To allow users to roughly grasp the number distribution of each cell, we set the size of each instance to be the same, and this size is determined so that all instances can be arranged in the cell with the largest number of instances. The background color of each cell varies with the number (mapping white to 0 and blue to the largest number).

2.2.2 Class-level View. Class-level view is enabled when a diagonal cell or class label is clicked in overall view. Since many classification results contain most of the instances in diagonal cells, this view

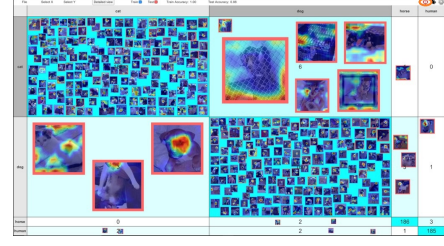


Figure 3: Activation mapping visualization.

attempts to display the instances in the clicked cell (i.e., diagonal cell) as large as possible. In this view, as shown in Fig. 1(b), the instances are enlarged and spread to fill the clicked cell, while the misclassified instances related to the focused class are represented as smaller than that in the focused cell but still visible and accessible. The instances in the other cells are reduced in size but the cells remain clickable.

2.2.3 Between-class View. Between-class view is enabled by clicking a non-diagonal cell, and it is used to perform between-class PA by observing instances in different cells simultaneously to support interactive inspection of common sources of errors (e.g., mislabeled data, anomaly data). Specifically, the 2×2 cells, including the clicked non-diagonal cell, are enlarged to show between-class confusion (e.g., when the *dog-cat* cell is clicked in overall view, the view changes as shown in Fig. 1(c), showing the confusion between *cat* and *dog* classes). The instances in focused cells then spread and fill the cells. The instances in the other cells are reduced in the same way as they are in the class-level view.

2.3 Practical Extensions

2.3.1 Activation mapping visualization. We implemented an extension that provides the results of activation mapping by directly overlaying them on each instance, which would lend insight into misclassification and show reasonable explanations for model behavior (Fig. 3). When the extension is enabled (by clicking an icon at the top-right of the window highlighted with orange circle), all of the instances are overlaid by their activation maps (our implementation uses Grad-CAM [6] for activation mapping because it is known to be a powerful method for localization and faithfulness to models).

2.3.2 Instance sorting/filtering. Another extension is an instance sorting/filtering function to facilitate the understanding of data distribution through rearranging the instances by metadata, referring to prior work on image visualization [2]. We provide data sorting/filtering inside each cell in two dimensions (horizontal and vertical axes) with several variables (e.g., image resolution, GT score: possibility score of the ground truth, and PD score: possibility score of the prediction). Fig. 4 shows instances sorted by the PD scores in the horizontal axis and filtered out at the range between 0.8 and 1.0 by adjusting the two horizontal sliders (highlighted with orange circles). This visualization, for example, can help users find the misclassified instances with high PD scores (e.g., highlighted with blue circles) quickly.

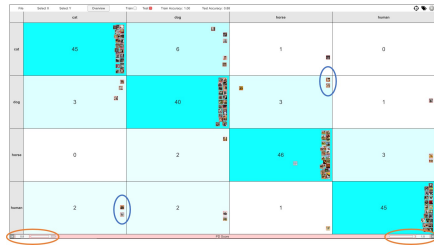


Figure 4: Instance sorting/filtering.

3 CONCLUSION AND FUTURE WORK

We proposed ConfusionLens, which aims to effectively assist ML users in achieving important image classification PA tasks with interactive focus+context visualization. Our prototype demonstrated that the interface can cover a dataset of ten classes with hundreds of instances. In addition, we implemented further extensions for practical PA tasks. Our future work includes clarifying the method’s scalability in terms of number of classes and instances and conducting a user study to investigate how our interface can support practical PA.

REFERENCES

- [1] Andreas Hinterreiter, Peter Ruch, Holger Stitz, Martin Ennemoser, Jurgen Bernard, Hendrik Strobelt, and Marc Streit. 2020. ConfusionFlow: A model-agnostic visualization for temporal analysis of classifier confusion. *IEEE Transactions on Visualization and Computer Graphics* (2020).
- [2] Xin Huang, Kazuki Takashima, Kazuyuki Fujita, and Yoshifumi Kitamura. 2018. Dynamic, Flexible and Multi-dimensional Visualization of Digital Photos and their Metadata. In *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces*. 405–408.
- [3] Michael Pühringer, Andreas Hinterreiter, and Marc Streit. 2020. InstanceFlow: Visualizing the Evolution of Classifier Confusion at the Instance Level. In *2020 IEEE Visualization Conference (VIS)*. 291–295. <https://doi.org/10.1109/VIS47514.2020.00065>
- [4] Ramana Rao and Stuart K Card. 1994. The Table Lens: Merging Graphical and Symbolic Representations in an Interactive Focus+Context Visualization for Tabular Information. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 318–322.
- [5] Donghao Ren, Saleema Amershi, Bongshin Lee, Jina Suh, and Jason D Williams. 2016. Squares: Supporting Interactive Performance Analysis for Multiclass Classifiers. *IEEE transactions on visualization and computer graphics* 23, 1 (2016), 61–70.
- [6] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. In *Proceedings of the IEEE international conference on computer vision*. 618–626.
- [7] Yalong Yang, Wenyu Xia, Fritz Lekschas, Carolina Nobre, Robert Krüger, and Hanspeter Pfister. 2022. The Pattern is in the Details: An Evaluation of Interaction Techniques for Locating, Searching, and Contextualizing Details in Multivariate Matrix Visualizations. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 84, 15 pages. <https://doi.org/10.1145/3491102.3517673>